NTNU

Department of Industrial Economics and Technology Management

Examination paper for TIØ4145 Corporate Finance

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Permitted examination support material: D - No printed or hand-written support material is allowed. A specific basic calculator is allowed.

Other information:

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Informasjon om trykking av eksamensoppgave

Originalen er:

1-sidig ☐ **2-sidig** ☐

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Date Signature

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Task 1 - Stocks and Bonds [20%]

a. Suppose you are trying to determine the interest rate sensitivity of two bonds. Bond 1 is a 12% coupon bond with yearly payouts, a 7-year maturity and a $1000 face value. Bond 2 is a 'zero-coupon' bond that pays $1000 after 7 year. The current interest rate is 12%.
   i. Determine the duration of each bond.
   ii. Which bond is more sensitive to interest rate changes? Explain.

b. The total value of a firm is 100 ($ million). The dividend policy of the firm is a 5% dividend yield; its debt policy is to be 75% equity financed. Investors expect a 16% return on stocks of firm A, and ask 4% interest on firm A's debt.
   i. What constant expected growth rate of the firm is in line with the numbers (assuming no new equity will be issued)?
   ii. What then are the expected earnings in year 1 and in year 5?

Task 2 - Financial Options [30%]

Consider a two-year European call option on Pineapple Inc. stock. The current stock price is 100 USD. In each of the next two years the stock price will either increase by 10% or decrease by 20%. The strike price of the call is 80 USD. The stock pays no dividends and the risk free interest rate is equal to 4%.

a. Using a two-period binomial model, calculate the price of this European call option.
b. Calculate the price of the same European call option using risk-neutral valuation. Show the steps explicitly.
c. Compare the outcomes in *a.* and *b.* and comment on the result.

Suppose that the Stock Exchange also listed an American call option with the same characteristics.

d. Did such a decision make sense if Pineapple Inc. was not expected to pay out dividends or repurchase shares in the foreseeable future? Explain your answer.
e. What is the price for this American option, and how does it compare to the price of the European option? Explain.
f. How would your answer in e. change if Pineapple Inc. decided to pay out dividends? Explain intuitively.

Task 3 - Capital Structure [10%]

As a financial consultant, you are given the following information about two companies, one levered and one unlevered:

	Unlevered	Levered
Before tax operating income	$10 000	$10 000
Interest on debt	0	2500
Cost of capital (WACC)		16%
Cost of Equity capital		
Cost of Debt capital		
Value of equity		22 500
Value of debt		
Value of fim		

Copy the table to your answer sheet and fill in the missing figures. Show the steps you used to calculate them. Assume that the corporate tax rate is 40% and all cash flows are perpetual.

Task 4- Portfolio Choice [25%]

Suppose that there are two stocks available in the market: stock A and stock B. The expected returns of these stocks are equal, and given by $E[R_A]=E[R_B]=0.16$. The standard deviation of stock A is equal to 0.4, and the standard deviation of stock B is 0.7. The stocks are positively correlated with the correlation coefficient equal to 0.18.

a. Find the expected return and volatility of the portfolio consisting of 20% of stock A and 80% of stock B.
b. Is this portfolio efficient? Explain.
c. Sketch the portfolios that one can create by combining stocks A and B for the following weights:

Weight Stock A	Weight Stock B
0%	100%
20%	80%
60%	40%
80%	20%
100%	0%

Illustrate the efficient frontier.

d. What is the optimal portfolio choice for a typical risk-averse investor?
e. Explain intuitively what correlation coefficient represents.
f. How would your answer to question *d.* change if the correlation coefficient changed? Consider two extreme examples: when stocks A and B are perfectly positively correlated and when they are perfectly negatively correlated.

Task 5 - Pricing Theories [15%]

In the discussions of some of the models in this course, we relied on the following type of argument: "If two investment strategies have the same payoff tomorrow, they must have the same price today."

a. What did we call this statement in the course? State the assumptions that are necessary for this statement to hold.

b. For which of the following theories, shortly explaining each theory and explain which of the following theories relies explicitly on no-arbitrage arguments.
   i. The Put Call parity of option pricing.
   ii. The Miller Modigliani propositions for capital structure.
   iii. The CAPM.